Phytomedicine 145 (2025) 157055

Contents lists available at ScienceDirect

Phytomedicine

journal homepage: www.elsevier.com/locate/phymed

Original Article

Bupi Yishen formula improves chronic kidney disease by restoring renal energy metabolism and mitochondrial oxidative phosphorylation

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ARTICLE INFO

Keywords:
Chronic kidney disease
Bupi Yishen formula
Fatty acid oxidation
Glucose metabolism
Mitochondrial oxidative phosphorylation

ABSTRACT

Background: Bupi Yishen formula (BYF) is a traditional Chinese herbal mixture with proven efficacy in attenuating kidney function deterioration among patients with advanced chronic kidney disease (CKD), and improving renal fibrosis of CKD animal models. Previous studies have shown that BYF rehabilitates metabolic dysregulation under CKD condition, but its exact mechanism remains unclear.

Purpose: This study aimed to elucidate the therapeutic effect and its potential mechanism on regulating renal energy metabolism in CKD.

Methods: An adenine-induced CKD rat model was treated with two doses of BYF decoction (15 g/kg/day or 30 g/kg/day) and losartan (as the positive control) for 4 weeks. Lipidomic and transcriptomic analyses of kidney samples from CKD rats revealed the BYF-reversed different lipid species and gene expression profiles respectively, thereby identifying potential pharmacological mechanisms. Further *in vivo* and *in vitro* experiments, network analyses, and molecular docking was used to confirm the proposed mechanisms affected by BYF.

Results: BYF had a profound impact on alleviating renal impairment and profibrotic phenotypes in CKD rats. Lipid profiling of kidneys from CKD rats showed that the dysmetabolism of glycerophospholipids, sphingolipids, and glycerolipids was primarily influenced by BYF. Transcriptome analysis of CKD rats identified renal energy

Abbreviations: Acadvl, Acyl-CoA dehydrogenase, very long chain; α-SMA, alpha smooth muscle actin; ATP5A, ATP synthase alpha-subunit; BYF, Bupi Yishen formula; Cat, Catalase; CerG3GNAc1, Ceramide-GlcNAc1; CKD, Chronic kidney disease; CL, Cardiolipin; CLYBL, Citramalyl-CoA lyase; COL1A1, Collagen I; COX4I2, Cytochrome c oxidase subunit 4I2; Cyp4a2, Cytochrome P450 family 4 subfamily A member 2; DEGs, Differentially expressed genes; DG, Diacylglycerol; Dnm1l, Dynamin 1-like; ECAR, Extracellular acidification rate; Eci3, Enoyl-coenzyme A delta isomerase 3; ETC, Electron transport chain; FAO, Fatty acid oxidation; Fbp1, Fructose-1,6-bisphosphatase 1; FC, Fold change; FN1, Fibronectin 1; Galm, Galactose mutarotase. G6pd, Glucose-6-phosphate dehydrogenase; Havcr1, Hepatitis a virus cellular receptor 1; Hex1Cer, Hexosylceramide; Hif1a, Hypoxia-inducible factor 1 subunit alpha; HK2, Hexokinase 2; Khk, Ketohexokinase; Lcn2, Lipocalin2; LPC, Lysophosphatidylcholine; Mfn2, Mitofusin 2; Mmp9, Matrix metalloproteinase 9; Ndufa11, NADH: Ubiquinone Oxidoreductase Subunit A11; Ndufs7, NADH: Ubiquinone Oxidoreductase Core Subunit S7; Notch1, Notch receptor 1; Nox4, NADPH oxidase 4; NRF1, Nuclear respiratory factor 1; OCR, Oxygen consumption rate; OXPHOS, Oxidative phosphorylation; PAS, Periodic acid-Schiff; PC, Phosphatidylcholines; Pc, Pyruvate carboxylase; PCA, Principal component analysis; Pdgfb, Platelet derived growth factor subunit B; PE, Phosphatidylethanolamine; PG, Phosphatidylglycerol; PI, Phosphatidylinositol; PKM2, Pyruvate kinase M2; PPP, Pentose phosphate pathway; PS, Phosphatidylserine; qRT-PCR, Quantitative reverse transcription polymerase chain reaction; RFUs, Relative fluorescence units; ROS, Reactive oxygen species; Scp2, Sterol carrier protein 2; SDH, Succinate dehydrogenase; SDHA, Succinate dehydrogenase complex flavoprotein subunit A; SDS-PAGE, Sodium dodecyl sulfate polyacrylamide gel electrophoresis; SM, Sphingomyelin; Sod1, Superoxide dismutase 1; ST, Sterol; TEM, Transmission electron microscopy; Tfam, Mitochondrial transcription factor A; TG, Triglyceride; Tkfc, Triokinase and FMN Cyclase; UHPLC-MS, Ultra-high-performance liquid chromatography-mass spectrometry; UQCRQ, Ubiquinol-Cytochrome C Reductase Complex III Subunit VII; VIP, Variable importance in projection; 6Pgd, 6-phosphogluconate dehydrogenase.

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https://doi.org/10.1016/j.phymed.2025.157055
Received 21 February 2025; Received in revised form 24 June 2025; Accepted 3 July 2025
Available online 3 July 2025

metabolism (including fatty acid oxidation [FAO], glucose metabolism) and mitochondrial oxidative phosphorylation (OXPHOS) as the key dysregulated pathways, which were reversed by BYF. Further experiments confirmed that BYF partially restored defective FAO, dysregulated glucose metabolism, and impaired mitochondrial OXPHOS in the kidneys of CKD rats and TGFβ1-induced human tubule HK-2 cells. Besides, network analyses combined with molecular docking demonstrated a strong binding effect of BYF's core compounds on key targets related to energy metabolism.

Conclusions: These results suggest that BYF protects against CKD by restoring renal energy homeostasis and mitochondrial OXPHOS, offering potential as an alternative therapy for renal fibrosis inherent to CKD.

Introduction

Chronic kidney disease (CKD) is a pervasive public health issue associated with increasing morbidity and mortality. It is estimated that CKD affects about 10 % of the global population (Kalantar-Zadeh et al., 2021) and has a prevalence of 8.2 % in China (Wang et al., 2023a). CKD is projected to become the fifth leading cause of death globally by 2040 (Foreman et al., 2018). Renal fibrosis is a common histopathological hallmark of CKD and serves as the primary pathological precursor to end-stage renal disease. Therefore, preventing renal fibrosis is crucial in determining the progression of CKD (Humphreys, 2018). However, current anti-fibrotic treatments are unsatisfactory, suggesting the urgent need for more effective therapeutic strategies in the context of CKD.

Metabolic dysregulation is critical for the fibrotic pathology and advancement of CKD (van der Rijt et al., 2022). CKD has been linked to aberrant lipid metabolism. For instance, lipid uptake and lipogenesis contribute to intracellular lipid accumulation, ultimately leading to renal dysfunction and fibrosis (Baek et al., 2022). However, Kang's landmark study indicated that defective fatty acid oxidation (FAO) in CKD patients and fibrotic mouse models is critical for secondary lipid accumulation and renal fibrosis (Kang et al., 2015). Due to the kidney's high energy requirements, particularly the proximal tubular epithelial cells (TECs) that preferentially use fatty acids as their main energy sources, primarily supported by FAO and mitochondrial oxidative phosphorylation (OXPHOS), to maintain tubular reabsorption and solute filtration (Liu et al., 2018). Mitochondria are recognized as the main sites for FAO, while peroxisomal FAO also contributes to redox homeostasis and collaborates with mitochondria to produce energy (Silva Barbosa et al., 2024). Deficits in FAO and mitochondrial OXPHOS can disrupt cellular metabolism, leading to lipotoxicity and a pathogenic fibrotic phenotype (Yuan et al., 2021).

Recent studies have reported that kidneys respond to injury through flexible metabolic adaptations, often shifting from FAO to anaerobic glycolysis. Enhanced glycolysis is essential for responding quickly to injury, compensating for energy deficits, and supporting rapid biosynthesis in the early stages of CKD. This shift facilitates cell proliferation and repair processes (Basso et al., 2021; Chen et al., 2023; Chrysopoulou and Rinschen, 2024). However, persistent activation of glycolysis can promote the progression of tubulointerstitial fibrosis (Ding et al., 2017; van der Rijt et al., 2022). Besides, gluconeogenesis, an enzymatic pathway opposite to glycolysis also plays a significant role in the pathophysiology of CKD (Verissimo et al., 2022). Suppressed gluconeogenesis and enhanced glycolysis can lead to impaired lactate clearance. Lactate is not only the main substrate for gluconeogenesis and a byproduct of glycolysis but its accumulation can provoke inflammation, nephrotoxicity, and accelerate CKD progression (Kierans and Taylor, 2024). Altogether, abnormal metabolism of lipids, glucose, and mitochondria is closely related to renal fibrosis, suggesting that correcting metabolic dysregulation may be crucial for treating CKD.

Traditional Chinese medicine is recognized as an alternative therapy for CKD treatment (Liu et al., 2022b; Zhong et al., 2015). As a patented herbal formula, Bupi Yishen formula (BYF) has been clinically proven effective in delaying CKD progression, and its inhibitory effects on renal fibrosis have been well established in our previous studies (Liu et al., 2022a, 2023; Mo et al., 2021; Zhang et al., 2021). These studies have shown that elevated free fatty acids, abnormal lipid accumulation, and increased pro-apoptotic phenotypes is observed in the fibrotic kidneys of CKD rat models, all of which are reversed by BYF treatment (Liu et al., 2023). These findings suggest that BYF exhibits anti-inflammatory, anti-fibrotic, and nephroprotective properties, possibly through the restoration of mitochondrial function and the normalization of lipid dysmetabolism. However, the exact mechanisms by which BYF protects against kidney energy abnormalities remain unclear.

In the present study, a comprehensive metabolic atlas of the kidneys in CKD rats was generated by integrating lipidomic and transcriptomic analyses. We also conducted experimental validation and molecular docking analysis to confirm the proposed mechanisms affected by BYF. Our results suggest that BYF offers renal protection in CKD, likely through the restoration of disrupted fatty acid metabolism, glucose metabolism, and mitochondrial OXPHOS in the kidneys, thus providing a compelling experimental basis for the clinical application of BYF in CKD treatment.

Methods

Animal study

This experiment was approved by the Ethical Review Committee of the Second Clinical Medical College of Guangzhou University of Traditional Chinese Medicine (No. 2023,054–2, trial period from June 1, 2023, to June 30, 2025). Sprague-Dawley (SD) male rats were housed in a specific-pathogen-free (SPF) facility, in compliance with international guidelines for experimental animal care. The rats were obtained from the Guangdong Provincial Laboratory Animal Center. All rats (8 weeks old, approximately 180–200 g) were randomly divided into 5 groups ($n = 6$) after a 1-week adaptation period.

CKD was induced in the rats with adenine (Sigma-Aldrich, St. Louis, MO, USA) to establish a CKD model. Two hours after adenine treatment, CKD rats were administered with two different dosages of BYF via oral gavage, once daily. Losartan was used as a positive control, as described previously (Liu et al., 2023). The detailed groupings were as follows: (a) Control group: normal diet and water intake. (b) CKD group: oral gavage of adenine (200 mg/kg/day). (c) CKD/BYF-l group: BYF decoction (15 g/kg/day). (d) CKD/BYF-H group: BYF decoction (30 g/kg/day). (e) CKD/Los group: losartan (20 mg/kg/day). Rat body weights were recorded weekly. The rats were euthanized after 4 weeks, and tissue specimens were collected for subsequent studies.

Identification of BYF chemical ingredients by ultra-high-performance liquid chromatography-mass spectrometry (UHPLC-MS)

The preparation of BYF decoction was described in a previous study (Liu et al., 2023), and detailed information of the 8 herbs contained in BYF decoction is shown in **Table S1.** UHPLC-MS was used to analyze the chemical composition of the BYF extracts and to quantify the principal components. Through chemical standardization, the chemical analysis of BYF extracts served as a quality control measure for the subsequent animal experiments.

Blood analysis

Blood samples were collected from the rats after a 12-hour fasting period. Serum creatinine and urea levels were measured in 50 μl of serum from each rat at the Laboratory Department of Guangdong Provincial Hospital of Traditional Chinese Medicine, University City Hospital (Guangzhou, China), using a Roche automatic biochemical analyzer (Hitachi, 7180, Tokyo, Japan).

Histological analysis

Fresh kidney samples were harvested, bisected, and fixed in 4 % formaldehyde (pH 7.2). The samples were then embedded in paraffin and sectioned into 4 μm thick slices. Periodic acid-Schiff (PAS) staining, Masson's trichrome staining, as well as Sirius red staining were carried out according to previously established protocols (Liu et al., 2022a; Zhang et al., 2021). For immunohistochemical staining, kidney sections were incubated with primary antibodies, including anti-α-SMA (1:100, #19,245, CST), anti-Fibronectin (1:100, #26,836, CST), and anti-collagen I (1:100, #72,026, CST) overnight at 4 °C. Subsequently the sections were incubated with HRP-conjugated secondary antibodies (Millipore, MA, USA) and visualized using the 3,3′-Diaminobenzidine substrate (DAB) substrate. Micrographs of the stained kidney sections were captured using a light microscope (Zeiss Imager A2, Germany) and analyzed using ImageJ (NIH, Bethesda, MD, USA).

Lipidomics analyses

A CSH C18 column (1.7 μm, 2.1 mm × 100 mm, Waters) was used for reversed-phase chromatographic separation. Lipid extracts were redissolved in 200 μl of 90 % isopropanol/acetonitrile and vortexed at 14,000 rpm for 15 min. A 3 μl sample was injected. Solvent A consisted of acetonitrile-water (6:4, v/v), and solvent B was acetonitrile-isopropanol (1:9, v/v), both containing 0.1 % formic acid and 0.1 mM ammonium, respectively. The initial mobile phase was 30 % solvent B at a flow rate of 300 μl/min for 2 min, followed by a linear increase to 100 % solvent B over 23 min. The mobile phase was then equilibrated at 5 % solvent B for 10 min. Mass spectra were acquired using a Q-Exactive Plus in both positive and negative ion modes. Lipid species were identified using the Lipid Search database. The mass tolerance for both precursors and fragments was set to 5 ppm. The raw data of Lipidomic profile was listed in **Table S2.** The quality control base peak chromatogram in both positive and negative ion modes is shown in **Fig. S1 A and B.**

Transcriptomic analyses

Kidney RNA samples from the control, CKD, and CKD/BYF-H groups were sent to Novogene Biotech (Beijing, China) for library construction and transcriptome sequencing using the Illumina Novaseq 6000 platform. Differentially expressed genes (DEGs) between the groups were compared using DESeq2 software (R.1.16.1). The Benjamini and Hochberg methods were applied to adjust the *P*-values, and the absolute value of the log2 (fold change [FC]) was used as the threshold for significant differential expression. Enrichment analysis was performed using clusterProfiler (R.3.4.4). Correlation analysis was carried out with WGCNA 1.69 (for correlation calculation) and visualized using Corrplot 0.92 (for plotting). The raw data of transcriptomic profile was presented in **Table S3.**

Transmission electron microscopy (TEM)

Renal specimens were sectioned into small pieces and fixed in 2.5 % glutaraldehyde for 12 h, followed by post-fixation with 1 % osmium tetroxide. The fixed sections were dehydrated in graded alcohol, permeated with acetone, and embedded in epoxy resin. Ultrathin sections (70–90 nm) were stained with uranyl acetate and lead citrate, and then visualized under an electron microscope (H-600, Hitachi, Tokyo, Japan).

ROS assessment

A total of 50 mg of fresh kidney tissue samples were accurately weighed for the assay. Using the O08 reactive oxygen species (ROS) probe, the intensity of red fluorescence is proportional to the ROS level within the kidney tissue. The intensity of ROS in the tissue is expressed as relative fluorescence units (RFUs) per milligram of protein concentration using the ROS assay kit (BB-460,522, BBoxiProbe, China). The measurement method followed the instructions provided in the kit manual.

Renal ATP measurement

Fresh kidney tissue samples and human tubule HK-2 cells were lysed in the provided lysis buffer from the kit to fully release intracellular target components, thereby providing suitable samples for subsequent experimental procedures. The ATP concentration in the sample was determined by mixing 20 μl of the supernatant with 100 μl of luciferase reagent. The luciferase in the reagent catalyzed the reaction between ATP and luciferin to produce fluorescence, with the fluorescence intensity being directly proportional to the ATP concentration. An ATP standard curve was constructed using a series of known concentrations of standards. The measured ATP was presented as nmol per milligram (nmol/mg) of protein. The Bioluminescence Kit (S0026, Beyotime, China) was used to measure the ATP concentration.

Succinate dehydrogenase (SDH) assessment

SDH, known as electron transport chain (ETC) Complex II, is involved in OXPHOS. The tissue is accurately weighed to 0.1 g, and the reagent is added for preparation. One unit of enzyme activity is defined as the consumption of 1 nmol of 2,6-dichlorophenol indophenol per minute per mg of tissue protein in the reaction system. The procedure details are provided in the kit instructions (BC0955, Solarbio, China).

Mitochondrial complex I activity assessment

The enzymatic activity of mitochondrial OXPHOS Complex I, the largest protein complex in the mitochondrial inner membrane, was determined using the Complex I Enzyme Activity Assay Kit (BC0515, Solarbio, China). Briefly, kidney tissues were isolated, and appropriate concentrations of protein were prepared separately using the lysis kit. The samples were then processed in 96-well plates. One unit of enzyme activity was defined as the consumption of 1 nmol of NADH per minute per mg of tissue protein.

Lactate content assessment

The lactate content is an important index for evaluating glucose metabolism and aerobic metabolism. Approximately 0.1 g of tissue was weighed, and the measurement was performed according to the manual instructions (BC2235, Solarbio, China), with a characterized absorption peak at 570 nm.

Quantitative reverse transcription polymerase chain reaction (qRT-PCR)

Total RNA was extracted from kidney tissue and HK-2 cells using Trizol (R401–01, Vazyme Biotech, Nanjing, China). First-strand cDNA was synthesized using the RT reagent kit (R211–01, Vazyme Biotech, Nanjing, China). Real-time PCR was performed using the SYBR Premix Ex kit (Q226–01, Vazyme Biotech, Nanjing, China) on a ROCHE 96-well qRT-PCR system. Relative transcript levels were calculated using the $2^{-\Delta\Delta Ct}$ method, with β-actin as the endogenous control. PCR primers are

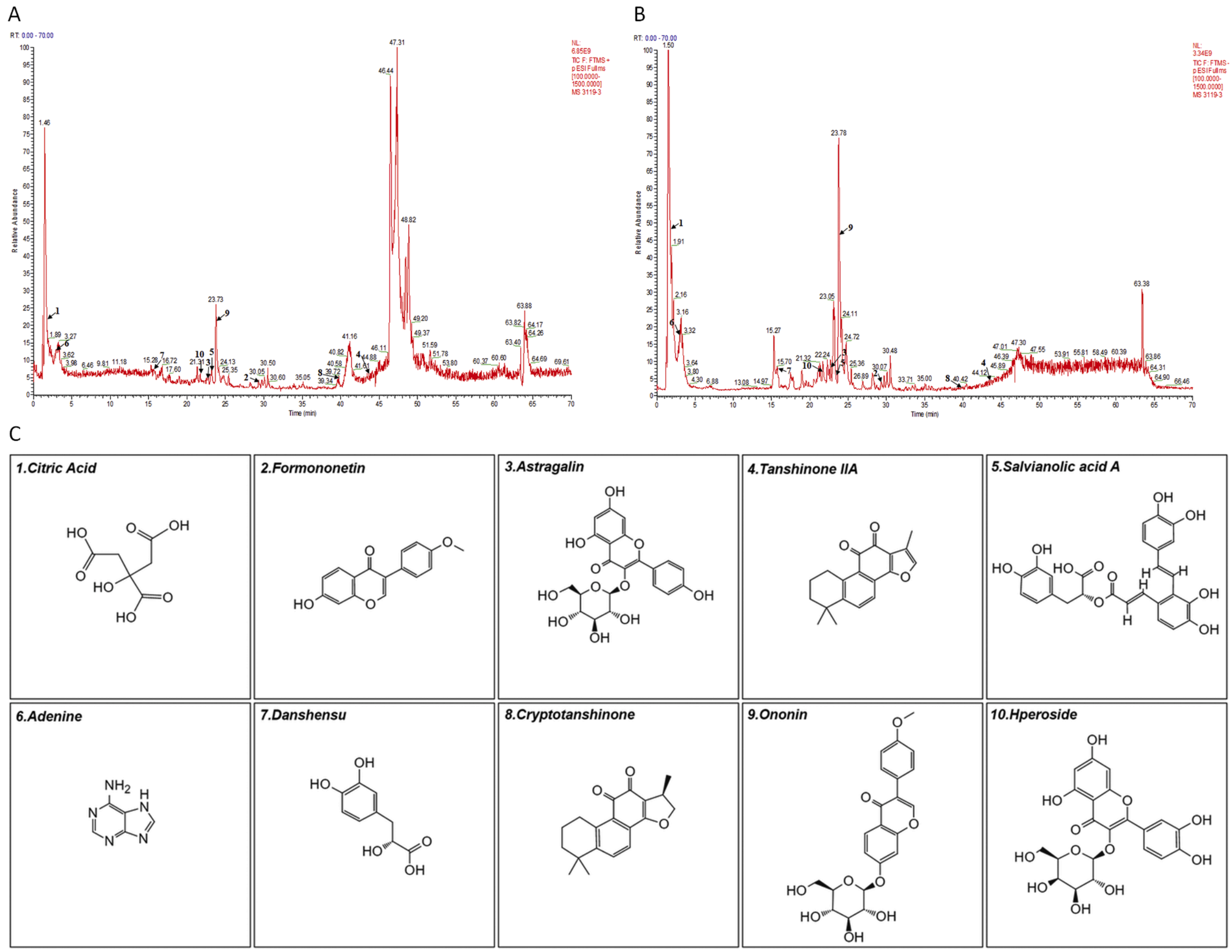

Fig. 1. BYF preparation. (A) TIC of BYF decoction in positive ion mode. (B) TIC of BYF decoction in negative ion mode. (C) The chemical structures of the main identified ingredients from BYF decoction. TIC, Total ion chromatogram.

listed in **Table S4**.

Western blotting

Renal cortex preparations were homogenized in RIPA buffer (#9806, CST, USA), and proteins were transferred from sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) gels to membranes. The membranes were blocked with 5 % skim milk and then incubated with primary antibodies against ATP synthase alpha-subunit (ATP5A1, #18,023, CST), Nuclear respiratory factor 1 (NRF1, #46,743, CST), β-actin (Sigma-Aldrich, St. Louis, MO, USA), Pyruvate kinase isozyme type M2 (PKM2, #4053, CST), Hexokinase 2 (HK2, #2867, CST). Results were analyzed semi-quantitatively using ImageJ.

Target prediction and molecular docking

Based on the chemical compounds of BYF identified by UHPLC-MS, potential targets of BYF were predicted using the Traditional Chinese Medicine Database (http://herb.ac.cn/). These targets were then intersected with BYF-reversed DEGs from the transcriptome profile to identify common targets. A network was constructed using Cytoscape 3.6.1, and key network topology parameters of the candidate components and selected targets were calculated. Target protein structures were downloaded from online databases (https://www.rcsb.org/) and (http://alphafold.com/), and stored in 3D structural PDB format.

The structures of small-molecular active components of BYF were downloaded from the PubChem database (https://pubchem.ncbi.nlm.nih.gov/). AutoDock was used to conduct docking between the target protein and the active ligand, to investigate their binding affinity. Docking results were visualized via Pymol. Molecular docking was repeated at least three times, after which the binding energies and predicted binding values (pKi in μM) of BYF'core components and energy metabolism-associated protein targets were calculated. The results are expressed as the mean ± standard deviation.

Cell culture

Routine cultures of HK-2 cells (CL-01,109, Pricella, China) were carried out in DMEM/F12 culture medium supplemented with 10 % fetal bovine serum, complying with the previously described protocols (Liu et al., 2022a). The cells were divided into the following groups: (a) Control group: cultured in normal medium. (b) TGFβ1 group: cultured in medium containing 10 ng/ml TGFβ1 (100–21–1MG, Pepro Tech, USA). (c) TGFβ1+BYF group: cultured in medium containing 10 ng/ml TGFβ1 and 30 mg/ml BYF extract. (d) TGFβ1+Feno group: cultured in medium containing 10 ng/ml TGFβ1 and 50 μM fenofibrate (Tricor), a PPARα agonist that acts as a mitochondrial metabolic modulator (Pawlak et al., 2015). All interventions were applied for 24 h.

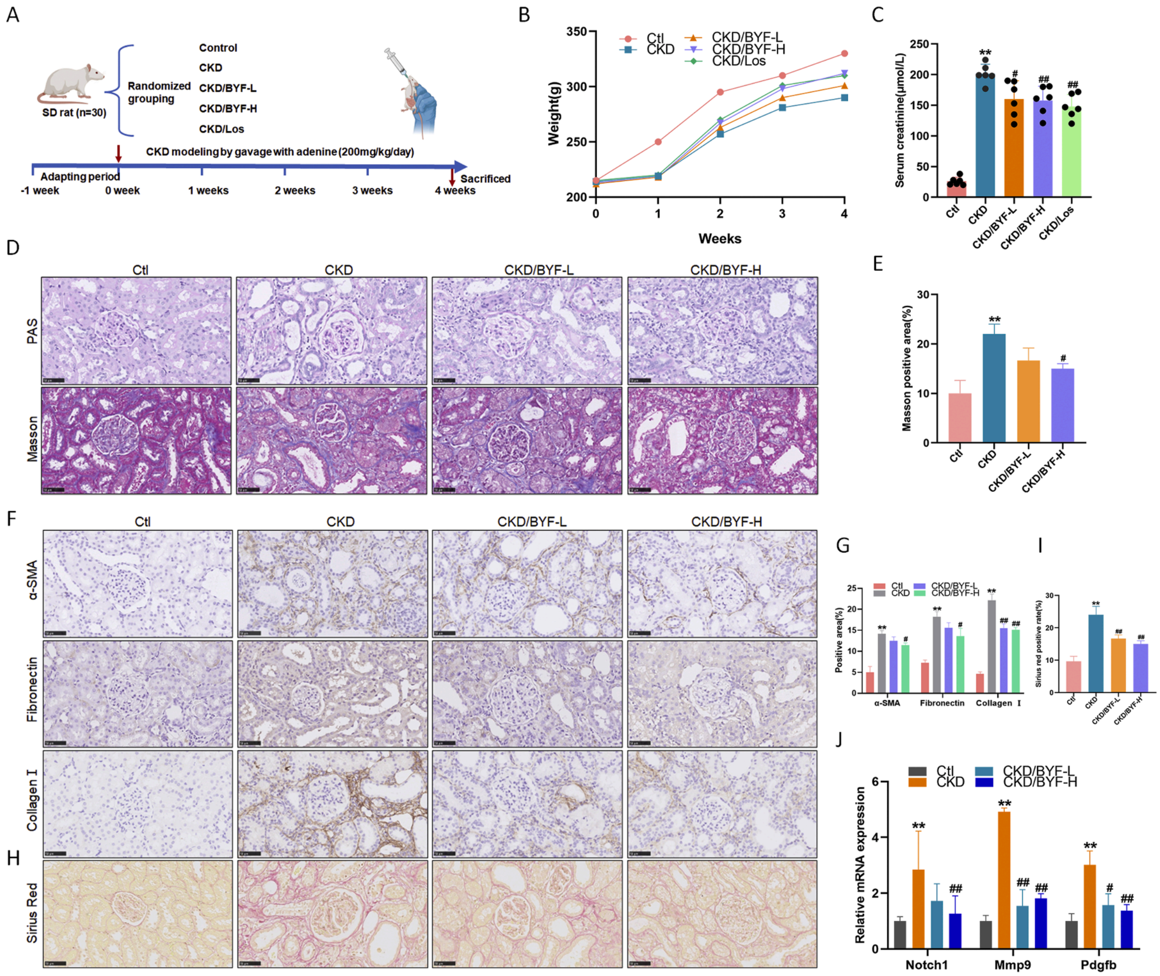

Fig. 2. Mitigation of renal impairment and histopathological changes in adenine-induced CKD rats by BYF. (A) Schematic diagram of the animal experimental procedure. (B) Average body weights at different time points. (C) Serum creatinine levels. (D-E) PAS and Masson staining of renal tissues (scale bar=50 μm) and semi-quantitative analysis of Masson staining. (F-G) Immunohistochemistry staining of α-SMA, Fibronectin, and Collagen I proteins (scale bar=50 μm) and semi-quantitative analysis. (H-I) Sirius Red staining of renal tissues (scale bar=50 μm) and semi-quantitative analysis of Sirius Red staining. **(J)** mRNA expression of *Mmp9, Notch1*, and *Pdgfb*. All data are representative results from one of three independent experiments. The values represent the mean ± SD, $n = 6$ rats per group (**$p < 0.01$, *$p < 0.05$ compared with the Control group, ##$p < 0.01$, #$p < 0.05$ compared with the CKD group).

Measurements of oxygen consumption rate (OCR) and extracellular acidification rate (ECAR)

After the intervention, HK-2 cells were resuspended in the ECAR and OCR assay working solution (E-BC-F068/69, Elabscience, China), maintaining a cell concentration of 5×10^5 cells/ml according to the instructions. Following a 30-minute incubation at 37 °C, measurements were performed using a multifunctional microplate reader.

Statistical analyses

Data were expressed as mean ± SEM. Comparisons between three or more groups were performed using one-way analysis of variance (ANOVA). All statistical analyses were conducted with GraphPad Prism version 8.0 (GraphPad Software Inc, La Jolla, CA). *p*-values <0.05 and *p*-values <0.01 indicated statistical significance and high statistical significance, respectively. In this study, most experiments yielded similar results. To ensure consistency in statistical analysis and avoid the repetitive presentation of similar data, only one representative experiment result was presented in this article.

Results

BYF alleviated renal impairment and mitigated the profibrotic phenotype changes in CKD rats

Given the batch effects of herbal medicine, we first used UHPLC-MS to identify 94 chemical ingredients from the BYF decoction extracts (Fig. 1**A, B**), and characterized their major active components, including Astragalin, Danshensu, Salvianolic acid A, Quercetin, and Citric acid (Fig. 1**C** and **Table S5**). To assess the reno-protective effect of BYF, an adenine-induced CKD rat model was constructed and the rats were treated with two dosages of BYF simultaneously (Fig. 2**A)**. BYF effectively prevented the gradual decrease in body weight observed in CKD rats (Fig. 2**B**). Moreover, BYF significantly inhibited the elevation of serum creatinine in CKD rats (Fig. 2**C**). Histopathological analysis revealed that extensive tubular injury and marked tubulointerstitial fibrosis in CKD rats were obviously mitigated by BYF treatment (Fig. 2**D, E**). In addition, the protein expression of profibrotic markers, including alpha smooth muscle actin (α-SMA), Fibronectin, and Collagen I, as assessed by immunohistochemistry (Fig. 2**F, G**), and the collagen accumulation observed with Sirius red staining (Fig. 2**H, I),** were significantly elevated in the CKD rats, but were markedly reduced in the BYF-treated rats. Similarly, BYF treatment decreased the mRNA

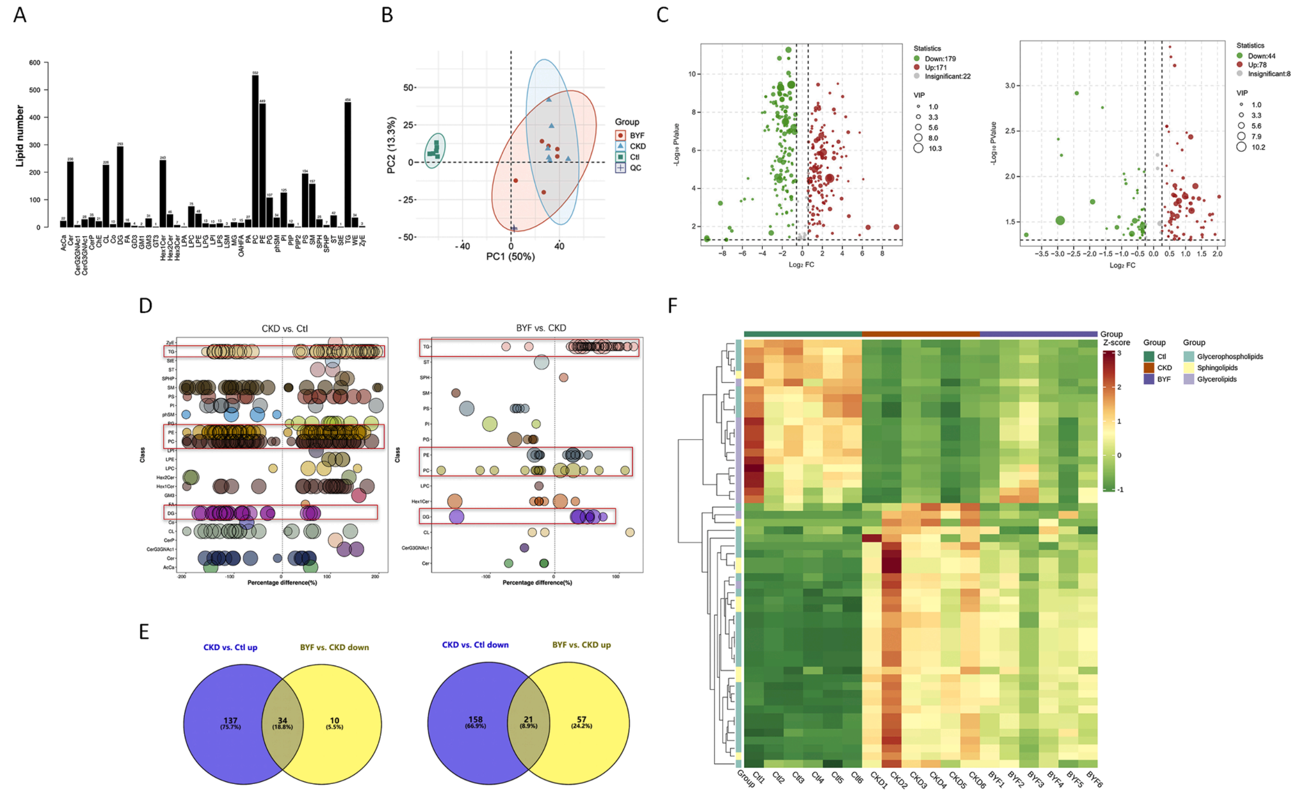

Fig. 3. Untargeted lipidomic profiles analysis. (A) Subgroups of lipids and number of lipid species. (B) PCA plots of 3652 lipid species. (C) Volcano plots of untargeted lipidomic analysis. Red dots represent up-regulated lipid species, green dots represent down-regulated lipid species, and gray dots represent lipids with no differences. Differential lipid species (VIP > 1, $p < 0.05$) are shown as volcano plots with -log10 vertically relative to log FC. The VIP value represents the importance projection value of the lipid species in the CKD model versus the control group, or the BYF group versus the CKD group. (D) Bubble plots comparing CKD versus control group and BYF versus CKD group. (E) Venn diagram of differential lipid species for each group. (F) Heatmap of clustering significantly differential lipid species and subclasses. VIP, variable importance in projection; FC, fold change.

expression of other profibrotic markers, such as matrix metalloproteinase 9 (*Mmp9*), notch receptor 1 (*Notch1*), and platelet derived growth factor subunit B (*Pdgfb*), in CKD rats (Fig. 2**J**). These results indicated that BYF effectively attenuated kidney lesions and profibrotic phenotypes in adenine-induced CKD rats.

Lipidomic profiles of the kidneys in CKD rats primarily indicated alterations in glycerophospholipids and sphingolipids and glycerolipids by BYF

Preliminary studies have shown that BYF improves aberrant fatty acid metabolism in the fibrotic kidneys of two CKD rat models (Liu et al., 2023). To comprehensively elucidate the effects of BYF on fatty acid dysmetabolism, the lipid profile of kidney samples collected from CKD rats was characterized using untargeted lipidome analysis. A total of 43 lipid classes and 3652 lipid species were identified across the three groups (Fig. 3**A**). Principal component analysis (PCA) scores demonstrated clear segregation between the control and CKD groups, with the BYF group moving closer to the control group (Fig. 3**B**). The volcano plots (Fig. 3**C**) revealed significantly changed lipid species in pairwise comparisons between the groups (variable importance in projection [VIP]>1, $|FC|\geq 1.5$, P-value < 0.05 for CKD vs. Control; VIP>1, $|FC|\geq 1.2$, P-value < 0.05 for BYF vs. CKD). Based on these differential lipid species, bubble plots identified significantly altered lipid subclasses between the groups, indicating that glycerophospholipids (e.g., Phosphatidylethanolamine [PE], Phosphatidylcholines [PC]), glycerolipids (e.g., Triglyceride [TG], Diacylglycerol [DG]), and sphingolipids (e.g., Hexosylceramide [Hex1Cer]) were changed in the CKD group and reversed by BYF treatment (Fig. 3**D**). The Venn diagram showed that 34 increased and 21 decreased content of lipid species in the CKD group were significantly reversed by BYF (Fig. 3**E**). Hierarchical clustering analysis visualized the lipid content of these 55 differential lipids and the representative subclasses (Fig. 3**F and Table S6**).

Further analysis revealed that a significant portion of glycerophospholipids (PE, PC, Phosphatidylserine [PS], Phosphatidylinositol [PI], Phosphatidylglycerol [PG], Cardiolipin [CL], and Lysophosphatidylcholine [LPC]) and sphingolipids (Hex1Cer, Sphingomyelin [SM], Sterol [ST], and Ceramide-GlcNAc1 [CerG3GNAc1]) were increased in the CKD group, but were reduced in the BYF group (Fig. 4**A, B**). Conversely, most glycerolipids (TG, DG) and some glycerophospholipids (PE, PC) were decreased in the CKD group but elevated by BYF (Fig. 4**C, D**). Although PC and PE are part of the same lipid class, they may have different or even opposite biological functions. Glycerophospholipids are essential for cellular function, but specific forms of PE and PC are associated with inflammation and fibrosis, which may contribute to CKD progression (Watanabe et al., 2014). Additionally, sphingolipid catabolic products can promote CKD and other inflammatory or fibrotic processes (Tanaka et al., 2022). These findings suggest that BYF may offer renal protection by inhibiting certain sphingolipids and glycerophospholipids while restoring glycerolipid metabolism balance under CKD conditions.

Transcriptomic analyses of the kidneys in CKD rats highlighted that BYF reversed the dysregulation of multidimensional energy metabolism

Excessive lipid accumulation is recognized as pathogenic, and disturbed lipid metabolism in the kidney accelerates CKD progression (Mitrofanova et al., 2023). Our previous studies have confirmed that BYF protects against CKD by inhibiting fibrogenesis and lipid dysmetabolism (Liu et al., 2023); however, the exact mechanisms remain

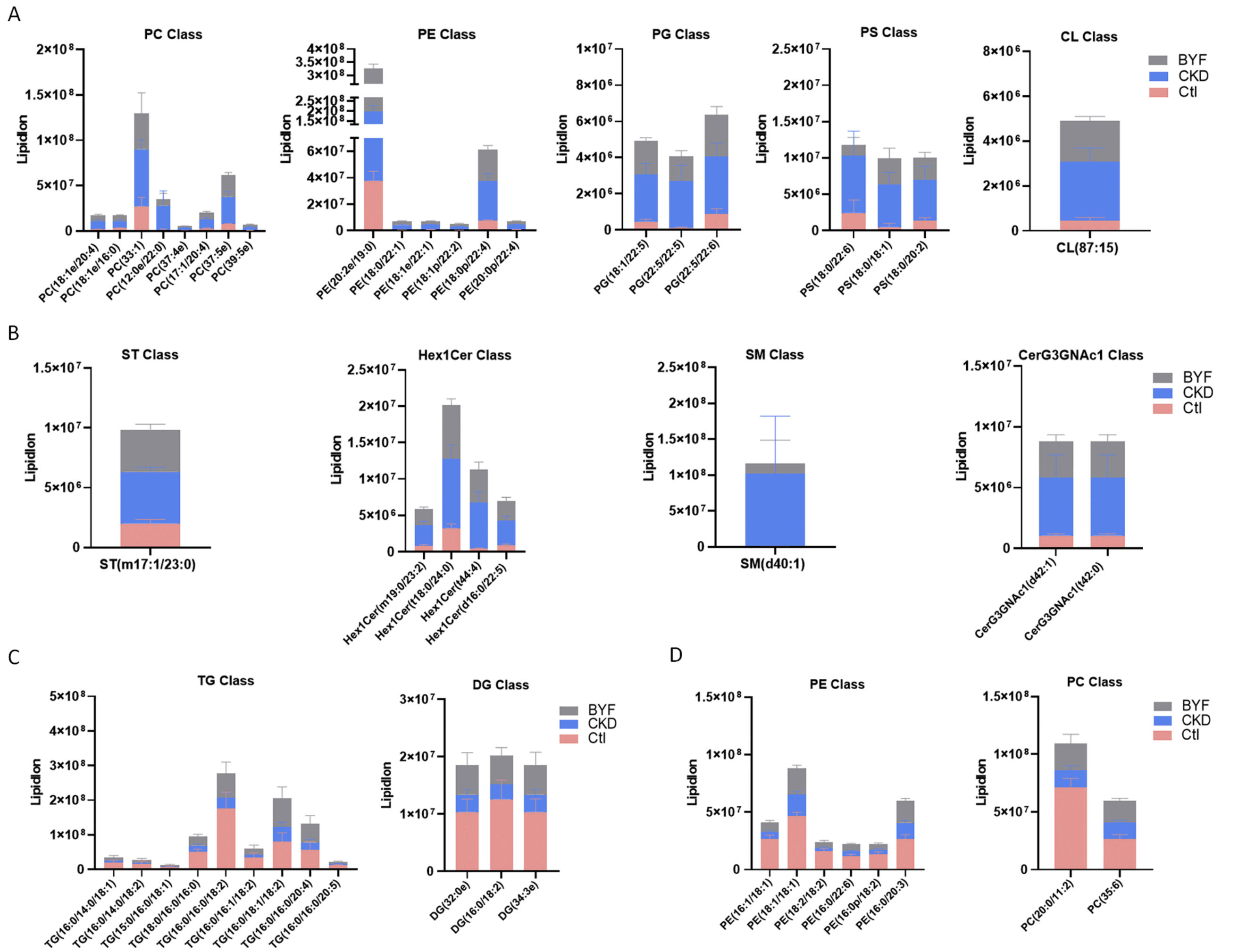

Fig. 4. Detailed differential lipid species among the groups. (A) Quantitative analysis of glycerophospholipids (PC, PE, PG, PS, CL). (B) Quantitative analysis of sphingolipids (ST, Hex1cer, SM, CerG3GNAc1). (C) Quantitative analysis of glycerolipids (TG, DG). (D) Quantitative analysis of glycerophospholipids (PE, PC).

unclear. In this study, a distinct trend of separation between the CKD and control groups was observed (Fig. 5**A)**, with a tendency for the BYF group to shift towards the control group. The volcano plots (Fig. 5**B**) highlighted DEGs between the groups ($|FC| \geq 2$, *P*-value < 0.05 for CKD vs. Control; $|FC| \geq 1.2$, *P*-value < 0.05 for BYF vs. CKD). The Venn diagram (Fig. 5**C**) showed that 198 upregulated and 246 downregulated DEGs in the CKD group were reversed by BYF. A heatmap of these 444 DEGs is presented in Fig. 5**D** Gene Ontology analysis showed that BYF-upregulated DEGs were mainly associated with renal development, metabolic processes, and mitochondrial OXPHOS (Fig. 5**E**). On the other hand, BYF-downregulated DEGs were predominantly enriched in pathways related to inflammation, cell cycle, cell migration, and MAPK cascade, all of which are linked to renal fibrogenesis and pathological lesions (Fig. 5**F**). Kyoto Encyclopedia of Genes and Genomes analysis further showed that BYF-increased DEGs mainly affected fatty acid metabolism, OXPHOS, and peroxisome functions (Fig. 5**G**). Conversely, BYF-decreased DEGs were associated with various pathological pathways related to renal inflammation and fibrosis, confirming the anti-inflammatory and anti-fibrotic effects of BYF, as established in previous studies (Liu et al., 2022a; Zhang et al., 2021) (Fig. 5**H**). Furthermore, we verified that the decreased mRNA expression of FAO-associated key genes (Enoyl-coenzyme A delta isomerase 3 [*Eci3*], Cytochrome P450 family 4 subfamily A member 2 [*Cyp4a2*], Acyl-CoA dehydrogenase, very long chain [*Acadvl*]), and mitochondrial homeostasis indicators (Mitofusin 2 [*Mfn2*], Mitochondrial transcription factor A [*Tfam*],*)* and increased Dynamin 1-like [*Dnm1l*], suggesting defective mitochondrial fission and fusion in CKD rats were reversed by BYF treatment (Fig. 5**I, J**). These findings indicated that BYF might exert a protective effect by restoring renal energy homeostasis through multiple metabolism pathways.

Moreover, the heatmap also depicted several energy-regulated genes involved in mitochondrial FAO (*Eci3, Cyp4a2, Acadvl*), peroxisomal antioxidation (Superoxide Dismutase 1 [*Sod1*], Catalase [*Cat*], Sterol Carrier Protein 2 [*Scp2*]), glucose metabolism (Ketonase [*Khk*], Triokinase and FMN Cyclase [*Tkfc*], Galactose mutarotase [*Galm*]), and OXPHOS (NADH: Ubiquinone Oxidoreductase Core Subunit S7, [*Ndufs7*], NADH: Ubiquinone Oxidoreductase Subunit A11, [*Ndufa11*], cytochrome c oxidase subunit 4I2 [*Cox4i2*]), which were reversed by BYF in the CKD condition (Fig. 6**A** and Table 1). To further investigate the link between altered lipid profiles and dysregulated metabolic pathways, Pearson correlation analysis between the selected genes related to energy metabolism (including *Eci3, Cyp4a2, Acadvl, Cat, Sod1, Scp2, Ndufs7, Ndufa11, Cox4i2, Khk, Tkfc, Galm*) and differential lipids in glycerophospholipid, sphingolipid, and glycerolipid subclasses was performed. A strong association was found between lipid subclasses and these energy-related genes or enzymes (Fig. 6**B**). Interestingly, these selected renal energy-regulated genes were positively associated with glycerolipid subclass changes (e.g., TG, DG), but negatively associated with glycerophospholipid subclass changes (e.g., PC, PG, PI, CL, LPC, PS, and PE) and sphingolipid subclass changes (e.g., ST, CerG3GNAc1, and Hex1Cer). These results indicated that lipid profile variations were closely associated with energy metabolism in the kidney. In conclusion,

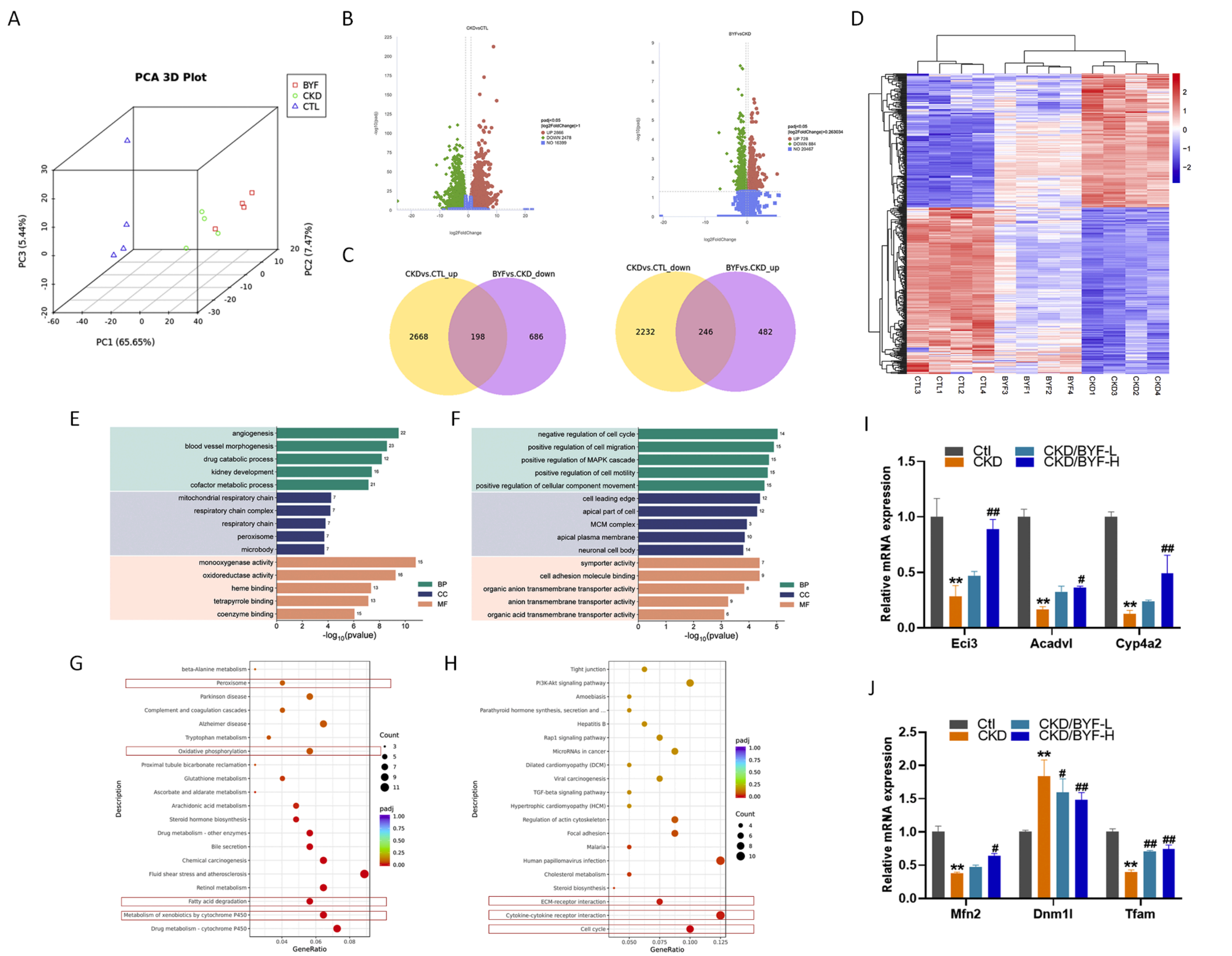

Fig. 5. Effect of BYF treatment on the renal transcriptome of adenine-induced CKD rats. (A) PCA score plot of the transcriptome of CKD rats among the groups. (B) Volcano plot of differential genes between the groups. Red or green dots indicate significantly up- or down-regulated genes; blue dots indicate non-significant DEGs. (C) Venn diagram of DEGs among the groups. (D) Heatmap cluster analysis of DEGs among the groups. (E) GO enrichment analysis of BYF-upregulated DEGs (including BP, MF, CC). (F) GO enrichment analysis of BYF-downregulated DEGs (including BP, MF, CC). (G) KEGG enrichment analysis of BYF-upregulated DEGs. (H) KEGG enrichment analysis of BYF-downregulated DEGs. (I) qPCR analysis of mitochondrial FAO indicators (*Eci3, Cyp4a2, Acadvl)* expression in each group. (J) qPCR analysis of mitochondrial functional indicators (*Mfn2, Dnm1l, Tfam*). (*$**p < 0.01$, *$*p < 0.05$ compared with the Control group, $\#\#p < 0.01$, $\#p < 0.05$ compared with the CKD group). BP, biological process; MF, molecular function; CC, cellular component; DEG, differentially expressed gene. GO, Gene Ontology; KEGG, Kyoto Encyclopedia of Genes and Genomes.

the integration of lipidomic and transcriptomic analysis suggested that BYF might exert a protective effect on CKD by restoring renal energetics and balancing lipid metabolism.

BYF partially restored the dysregulation of energy metabolism and improved impaired mitochondrial oxphos in the kidneys of CKD rats

Mitochondria are central to cellular metabolism, and their dysfunction is a key contributing factor in CKD (Kayhan et al., 2023). Structural mitochondrial damage in the kidneys of CKD rats, such as mitochondrial swelling, fragmentation, disruption of membrane integrity, as well as the absence or breakage of cristae, was partially ameliorated by BYF treatment (Fig. 7**A**). Additionally, OXPHOS, the primary pathway for energy production in the kidney, was compromised in CKD. Besides, the reduction in SDH (Complex II) content (Fig. 7**B**), the decreased enzymatic activity of respiratory chain complex I (Fig. 7**C**), and the down-regulation of ATP5A1 (a component of mitochondrial Complex V), and NRF1 (Wenz, 2013), which are key regulators of mitochondrial OXPHOS, were partially restored by BYF in the kidneys of CKD rats (Fig. 7**D, E**). As expected, BYF obviously alleviated the reduction in ATP levels, increased peroxisomal antioxidant-related key enzymes (*Sod1, Cat*), and reduced ROS accumulation in CKD rats (Fig. 7**F, G, H**). These results indicated that BYF might help restore mitochondrial function and improve ATP production by addressing OXPHOS and FAO deficiencies. When FAO and mitochondrial homeostasis are disrupted, glycolysis typically compensates. However, prolonged reliance on glycolysis can exacerbate the fibrotic process. In our study, we observed that BYF inhibited the elevated mRNA and protein levels of key glycolytic enzymes and activators, such as *Hk2, Pkm*, hypoxia-inducible factor 1 subunit alpha (*Hif1a*) (Fig. 7**I-K**), as well as the increased lactate content in CKD rats (Fig. 7**I**). Glycolysis activation is also associated with the pentose phosphate pathway (PPP), which may impair antioxidant capacity (Stincone et al., 2015) and lead to disordered gluconeogenesis, contributing to accelerated renal fibrosis (Verissimo et al., 2022). The decreased mRNA levels of key PPP enzymes (glucose-6-phosphate dehydrogenase [*G6pd*], 6-phosphogluconate dehydrogenase [*6Pgd*] and NADPH oxidase 4 [*Nox4*]) and gluconeogenesis enzymes (Pyruvate carboxylase [*Pc*], Fructose-1,6-bisphosphatase 1 [*Fbp1*]) among CKD rats were reversed by BYF treatment (Fig. 7**M, N**). In summary, BYF improves renal energy metabolism in CKD rats, likely through the

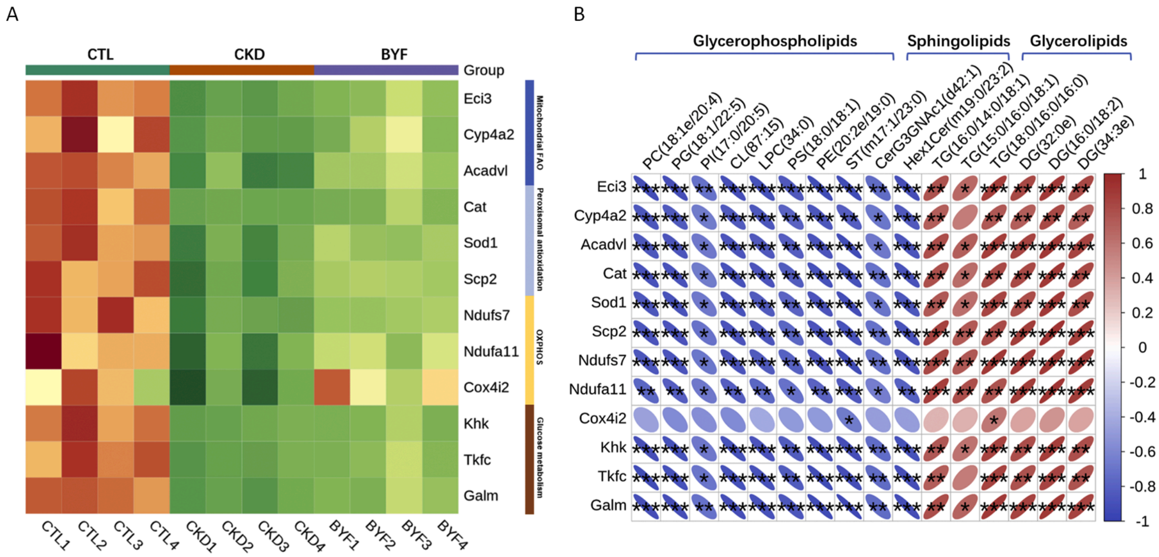

Fig. 6. Integration of transcriptomic and lipidomic analysis. (A) Heatmap of genes and enzymes related to FAO, glucose metabolism, and mitochondrial OXPHOS in each group. (B) Correlation analysis between selected differential lipid subclasses and genes related to metabolic pathways in each group. FAO, fatty acid oxidation; OXPHOS, oxidative phosphorylation.

restoration of FAO, glucose metabolism, and mitochondrial OXPHOS.

BYF significantly attenuated defective oxphos and suppressed enhanced glycolysis in TGFβ1-induced HK-2 cells

Renal tubule cells play a central role in energy production and are particularly vulnerable to metabolic disturbances in the kidney (Liu et al., 2018). The above results suggest that BYF has a regulatory effect on mitochondrial energy metabolism, thus we supplemented the mitochondrial metabolic modulator PPAR agonist (fenofibrate (Tricor), Feno) as a positive control in order to facilitate the confirmation of the role of BYF. In our study, BYF reduced the increased mRNA expression of tubular injury markers (Lipocalin2 [*LCN2*], Hepatitis a virus cellular receptor 1 [*HAVCR1*]) and fibrotic markers (Collagen I [*COL1A1*], Fibronectin 1, [*FN1*]) in TGFβ1-induced HK-2 cells (Fig. 8**A, B**). Additionally, BYF treatment helped restore the downregulated mRNA expression of key mitochondrial OXPHOS genes (Ubiquinol-Cytochrome C Reductase Complex III Subunit VII [*UQCRQ*], succinate dehydrogenase complex flavoprotein subunit A [*SDHA*], citramalyl-CoA lyas [*CLYBL*], *and COX4I2*) in TGFβ1-induced HK-2 cells (Fig. 8**C**). To gain further insight into the cellular metabolic changes, we performed FAO-related OCR assays (associated with OXPHOS) and ECAR assays (related to lactate production and glycolysis) in TGFβ1-induced HK-2 cells. BYF partially reversed the increase in ECAR (Fig. 8**D**), the decrease in OCR (Fig. 8**E**) and the reduction in ATP production (Fig. 8**F**) observed in these cells. Overall, BYF appeared to modulate renal tubule cellular dysmetabolism by inhibiting glycolysis and enhancing OXPHOS with effects similar to those of Feno, which, in turn, helped retard the progression of CKD.

Joint association of BYF core components with key targets of energy metabolism

Based on the 94 chemical ingredients identified from BYF using UHPLC-MS, a total of 1500 predicted targets were obtained from the Herb database (**Table S7**). The intersection of these 1500 predicted targets and the 444 BYF-reversed DEGs from the transcriptome profile resulted in 49 common targets implicated in BYF's therapeutic effects (**Fig. S2 A**). Besides, we constructed a network incorporating the 49 common targets and the 94 chemical components of BYF, and further identified 30 BYF' components were potentially interacted with those common targets (**Fig. S2 B** and **Table S8**). Finally, several key components (including Astragalin, Quercetin, Formononetin, Hyperoside, Tanshinone IIA, and Citric acid) and important targets related to energy metabolism (e.g., SOD, CAT, CLYBL, ACSS2, and COX4I2) were further screened for molecular docking analysis (Fig. 9). The docking details and parameters are presented in **Table S9.** These results reinforce the hypothesis of a joint association between BYF and renal energy metabolism in CKD.

Discussion

Our present study indicated that BYF primarily affects the metabolic abnormalities of glycerophospholipids, sphingolipids, and glycerolipids in the kidneys of CKD rats, as shown by untargeted lipidomic analysis. Further transcriptome profiling suggested that BYF mitigates renal energy dysmetabolism mainly by regulating fatty acid metabolism and mitochondrial OXPHOS. Moreover, experimental verification confirmed that BYF-mediated protection involves restoring renal energy by addressing defective FAO, disturbed glucose metabolism, and impaired mitochondrial OXPHOS. Network analysis and molecular docking further revealed the joint association between BYF's core components and multiple key targets related to energy metabolism. Thus, BYF may provide effective protection against CKD by restoring renal energetics and mitochondrial OXPHOS, as depicted in Fig. 10.

Disturbed lipid metabolism contributes to the fibrotic process in the kidney. Lipid accumulation is a crucial factor in renal fibrosis, and excessive lipid deposition and elevated free fatty acids in the fibrotic kidneys of CKD rats were observed in our previous study (Liu et al., 2023). Under CKD conditions, deciphering the mechanisms of lipid dysmetabolism is challenging due to its multiple structural and functional roles, as well as its involvement in various biological processes (Mitrofanova et al., 2023). Therefore, a comprehensive identification of differential lipids in renal lipid profiling affected by BYF in CKD rats may provide novel insights for CKD treatment. In the present study, the effect of BYF on renal lipid profiling alterations in adenine-induced CKD rats was preliminarily analyzed. Notably, elevated levels of circulating TGs are correlated with CKD, along with higher TG to high-density lipoprotein ratios, which are considered an increased risk for CKD (Weldegiorgis and Woodward, 2022). However, hemodialysis patients have been found to have lower total TG levels than healthy controls in certain cases (Bello et al., 2022). Furthermore, fatty acid dysmetabolism

Table 1
BYF reverses down-regulation of metabolic pathway-related key genes in CKD rats.

Gene symbol	CKD vs. Ctl			BYF vs. CKD			Functions related to metabolic pathways
	log2FC	p-value	p-adj	log2FC	p-value	p-adj	
Eci3	−2.271429706	7.2392767449106e-40	5.02080563228546e-38	0.76409395767199	0.000101477219997718	0.00486844633416295	Identification and characterization of Eci3, a murine kidney-specific delta3, delta2-enoyl-CoA isomerase (van Weeghel M et al., 2014, PMID: 24,344,334)
Cyp4a2	−3.768230551	3.41359048553593e-30	1.36131143904435e-28	1.98336512810148	0.000261480673011007	0.00837921811002762	LBP Protects Hepatocyte Mitochondrial Function Via the PPAR-CYP4A2 Signaling Pathway in a Rat Sepsis Model (Song Z et al., 2021, PMID:33,988,537)
Acadvl	−1.37063345	1.03809134572023e-15	1.46004001027015e-14	0.505280801295571	0.00209043996919725	0.0289043392461449	Novel ACADVL variants resulting in mitochondrial defects in long-chain acyl-CoA dehydrogenase deficiency (Chen T et al., 2020, PMID: 33,150,772)
Cat	−3.102060679	3.66E-87	3.18E-84	0.808096116	0.000908901	0.017791067	PEX13 prevents pexophagy by regulating ubiquitinated PEX5 and peroxisomal ROS (Demers et al., 2023, PMID: 36,541,703)
Sod1	−1.639153669	8.28E-18	1.38E-16	0.550760662	0.001639134	0.025177647	Mitophagy regulates mitochondrial network signaling, oxidative stress, and apoptosis during myoblast differentiation (Baechler et al., 2019, PMID: 30,859,901)
Scp2	−1.305259546	2.91915411223896e-14	3.645039009555e-13	0.470408246745714	0.00170068432903157	0.0256269342281214	SCP2 mediates the transport of lipid hydroperoxides to mitochondria in chondrocyte ferroptosis (Dai T et al., 2023, PMID: 37,422,468)
Ndufs7	−1.318912244	5.38110826979312e-16	7.77397543399094e-15	0.396923421054725	0.00170214150563765	0.0256269342281214	A novel mutation of the NDUFS7 gene leads to activation of a cryptic exon and impaired assembly of mitochondrial complex I in a patient with Leigh syndrome (Lebon S et al., 2007, PMID: 17,604,671)
Ndufa11	−1.067523862	1.0779699736816e-08	7.62264545113155e-08	0.456901436162148	0.00425772808522841	0.0445331478840479	Mitochondrial complex I deficiency caused by a deleterious NDUFA11 mutation (Berger I et al., 2008, PMID: 18,306,244)
Cox4i2	−1.895816389	0.00155710254049946	0.0049103882751632	1.82535188489805	0.000579509267027873	0.0136746626131698	Reprogramming of Mitochondrial Respiratory Chain Complex by Targeting SIRT3-COX4I2 Axis Attenuates Osteoarthritis Progression (Zhang Y et al., 2023, PMID: 36,683,245)
Khk	−3.562565803	3.26401826639147e-43	2.70475487685132e-41	0.997589039403585	0.00219657046123244	0.0297472942288992	KHK-A promotes fructose-dependent colorectal cancer liver metastasis by facilitating the phosphorylation and translocation of PKM2(Peng C et al., 2024, PMID: 39,027,256)
Tkfc	−2.546166449	5.60797721556994e-38	3.44063781603974e-36	0.774808945972332	0.00106959332900267	0.0195977011974648	Recent Progress on Fructose Metabolism-Chrebp, Fructolysis, and Polyol Pathway (Iizuka K., 2023, PMID: 37,049,617)
Galm	−1.65497056	5.51653144368076e-41	4.07712142451494e-39	0.427975981350451	0.00507777773239877	0.0494006220963722	Comprehensive Analyses of Glucose Metabolism in Glioma Reveal the Glioma-Promoting Effect of GALM (Xu J et al., 2022, PMID: 35,127,693)

Abbreviations: BYF, Bupi Yishen formula; CKD, Chronic kidney disease; FC, Fold change; p-adj, Adjusted p-value.

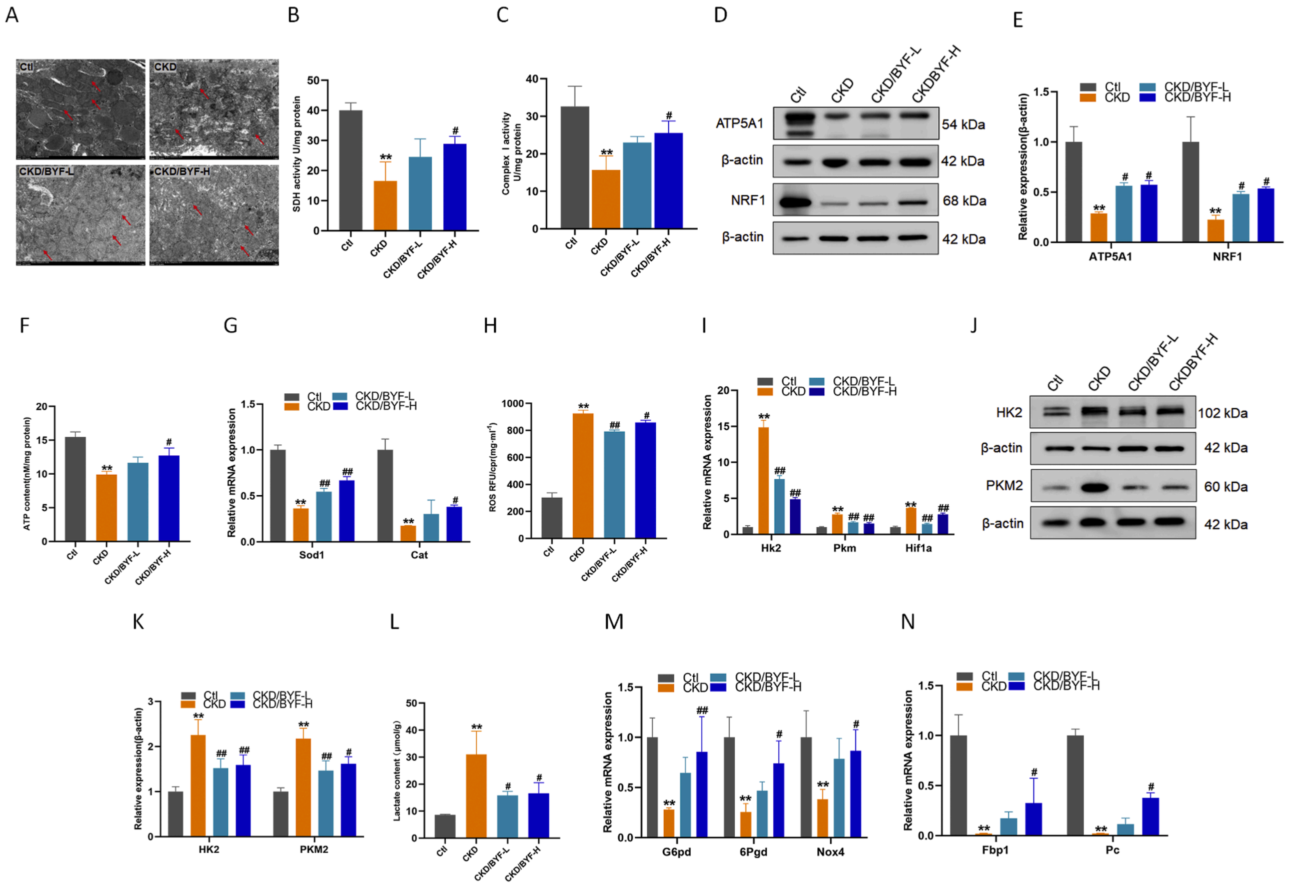

Fig. 7. Restoration of energy dysmetabolism in the kidneys of CKD rats following BYF treatment. (A) Representative electron micrographs of kidney tissue. Each image represents one group (scale bar=2 μm). (B) SDH content. (C) Mitochondrial respiratory chain complex I enzymatic activity. (D-E) WB and semi-quantitative analysis of OXPHOS functional indicators (ATP5A1, NRF1) in each group. (F) ATP content. (G) qPCR analysis of peroxisomal indicators (*Sod1, Cat*) expression in each group. (H) Renal tissue ROS production. (I) qPCR analysis of the glycolytic indicators (*Hk2, Pkm, Hif1a*). (J-K) WB and semi-quantitative analysis of glycolytic indicators (PKM2, HK2) in each group. (l) Renal tissue lactate content assay in each group. (M) qPCR analysis of the pentose phosphate pathway indicators (*G6pd, Nox4, 6Pgd*) expression in each group. (N) qPCR analysis of gluconeogenesis indicators (*Fbp1, Pc*) expression in each group. ($^{**}p < 0.01$, $^{*}p < 0.05$ compared with the Control group, $\#\#p < 0.01$, $\#p < 0.05$ compared with the CKD group). SDH, succinate dehydrogenase; ROS, reactive oxygen species.

due to renal fibrosis similarly affects TG synthesis, which in turn affects the synthesis of DGs. These reports imply that higher glycerolipid levels may contribute to CKD onset, but CKD or kidney fibrosis could also perturb glycerolipid synthesis. Our present data also reflect a downward trend in both DGs and TGs within the kidneys of CKD rats, with BYF partially recovering this alteration (Fig. 4**C**). Additionally, as a substrate for PC and PE synthesis, decreased TG abundance may reflect increased energy demand and active lipid synthesis (Cui and Liu, 2020). Correspondingly, we observed a decreasing trend in some glycerophospholipids (PE, PC) in CKD rats, which were reversed by BYF (Fig. 4**D**). Furthermore, mitochondrial membranes are rich in phospholipid species, mainly PC and PE (Baek et al., 2022), and contain lipid synthesis enzymes, actively contributing to the biosynthesis of phospholipids and phosphatidic acids. A decline in PE and PC may indicate disruption of the mitochondrial lipid membrane structure, which is an optimal stimulus for energy metabolism, thereby driving renal fibrosis.

Although glycerophospholipids are essential for the formation of membrane organelles, they can also be harmful in certain pathological conditions due to their structural complexity and diversity. PEs contribute to nephrotoxicity by promoting inflammation, oxidative stress, and lipid metabolism disorders. Higher levels of PCs and PEs have also been found in CKD cohort patients (Baek et al., 2022). Additionally, PI is a key component in activating the PI3K/Akt/mTOR signaling pathway. Consistently, we previously reported that BYF protects against renal fibrosis by suppressing the activation of this pathway (Liu et al., 2022a), and this effect may be related to the decline of PI. As for LPC, a group of phospholipids containing a fatty acid, it induces the pro-inflammatory mediator cyclooxygenase-2 (Mei et al., 2024), which triggers an inflammatory response and is involved in the fibrotic process. Moreover, CKD progression is also linked to sphingolipids, which play an important regulatory role, with ceramide being the core molecule in their synthesis (Wang et al., 2023b). Clinical studies have found that CKD patients typically exhibit elevated plasma ceramide levels (Shayman, 2018). Consistently, BYF significantly reduced the increased levels of most phospholipids and sphingolipids in the kidneys of CKD rats (Fig. 4**A, B**). Taken together, the modulatory effects of BYF on lipid dysmetabolism may be crucial in delaying CKD progression.

Excessive lipid accumulation promotes nephron fibrosis, and the main triggers of lipid disorders still need to be elucidated. In this study, we analyzed transcript data from kidney samples of CKD rats, revealing that BYF significantly improves two sets of differential genes: those related to inflammation and energy metabolism pathways (Fig. 5**G, H**). Differential lipids were strongly associated with metabolic pathway gene expression (e.g., Eci3, Cyp4a2, Acadvl, Cat, Sod1, Scp2, Ndufs7, Ndufa11, Cox4i2, Khk, Tkfc, and Galm) based on comprehensive omics analyses (Fig. 6**B**). We identified a link between the FAO/OXPHOS pathway and aberrant lipid metabolism in our current work. Given that we have already demonstrated BYF's protective effects against CKD through anti-inflammatory/anti-fibrotic mechanisms (Liu et al., 2022a; Mo et al., 2021; Zhang et al., 2021), several studies have highlighted the impact of dysregulated energy metabolism on the progression of CKD (Miguel et al., 2025; Singh, 2023; van der Rijt et al., 2022).

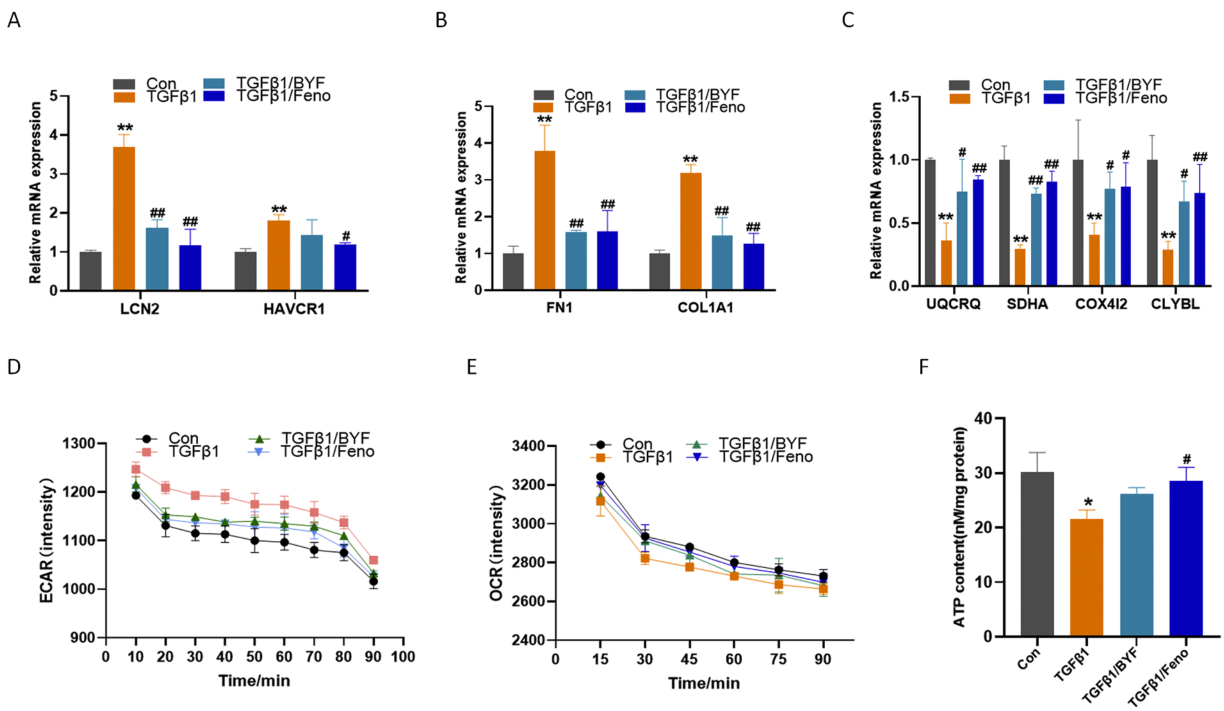

Fig. 8. Alteration of energy dysmetabolism in TGFβ1-induced HK-2 cells by BYF treatment. (A) qPCR analysis of the tubular injury indicators (*LCN2, HAVCR1*) expression in TGFβ1-induced HK-2 cells. (B) qPCR analysis of the fibrotic indicators (*COL1A1, FN1*) expression in TGFβ1-induced HK-2 cells. (C) qPCR analysis of the mitochondrial OXPHOS indicators (*UQCRQ, SDHA, CLYBL, and COX4I2*) expression in TGFβ1-induced HK-2 cells. (D) ECAR in TGFβ1-induced HK-2 cells. (E) OCR in TGFβ1-induced HK-2 cells. (F) ATP content in TGFβ1-induced HK-2 cells. (**$p < 0.01$, *$p < 0.05$ compared with the Control group, ##$p < 0.01$, #$p < 0.05$ compared with the TGFβ1 group). OXPHOS, oxidative phosphorylation; ECAR, Extracellular acidification rate; OCR, Oxygen Consumption Rate; Feno, fenofibrate (Tricor).

Understanding the interaction between energy metabolism and fibrosis can help unravel the mechanisms behind fibrosis formation and open up new avenues for the development of effective therapeutic interventions. Therefore, we will now discuss in detail the relationship between the dysfunction of FAO, glucose metabolism, and OXPHOS, focusing on the metabolic signature that drives CKD progression and restores energetic homeostasis.

Impairment of FAO often occurs in mitochondria and peroxisomes, playing a critical role in the pathogenesis of CKD. This is accompanied by the accumulation of free fatty acids, depolarization of the mitochondrial membrane potential, and loss of OXPHOS coupling (Lim et al., 2018; Singh, 2023). Peroxisomal FAO helps prevent the accumulation of toxic long-chain fatty acids in TECs, which is essential for counteracting lipotoxicity, and maintaining lipid metabolic homeostasis (Vasko, 2016). BYF partially restored catalase and other enzymes that neutralize ROS which are involved in peroxisomal redox homeostasis, likely preventing nephrotoxic damage and improving peroxisomal antioxidation (Fig. 7**G**). Mitochondrial homeostasis is a key determinant of renal energy maintenance (Bhargava and Schnellmann, 2017). While FAO and OXPHOS predominantly occur in mitochondria, peroxisomes do not directly produce energy but instead facilitate mitochondrial energy production by transferring short-chain FAO products to mitochondria (Silva Barbosa et al., 2024). However, progressive impairment of the mitochondrial ETC leads to FAO insufficiency, excessive ROS production, and activation of lipid peroxidation, which exacerbates mitochondrial damage and OXPHOS dysfunction, particularly in response to persistent inflammation and increased oxidative stress (Singh, 2023). In this study, we observed that BYF exerted a restorative effect on defective mitochondrial FAO and peroxisomal antioxidants (Fig. 5**I** and Fig. 7**G**).

Recent studies have demonstrated that fibrotic kidneys undergo metabolic reprogramming in response to injury, characterized by inhibited FAO and enhanced glycolysis. While the initial phase of activated glycolysis compensates for energy expenditure, there is growing evidence that glycolytic enzymes and the glycolytic end product lactate also contribute to inflammation and fibrosis (Kierans and Taylor, 2024). HK2 and PKM2 are involved in rate-limiting reactions during glycolysis, while HIF-1α triggers the transcription of glycolytic genes (Jha et al., 2015; Yang et al., 2023). An upward trend in these three key indicators of glycolysis, along with increased lactate levels in the kidneys of CKD rats, was reduced by BYF (Fig. 7**I-l**). Decreased lactate clearance suggests defective renal gluconeogenesis, which is a common feature of CKD (Verissimo et al., 2022). BYF likely modified renal gluconeogenesis deficiency and curtailed the sustained enhancement of glycolysis to inhibit CKD progression (Fig. 7**l** and Fig. 7**N**). Additionally, the PPP is a parallel glucose oxidation pathway that overlaps with glycolysis and is hyperactive in CKD, driven by sequential reactions of G6PD and 6PGD (Stincone et al., 2015). Reinstating the PPP suggests that BYF may play a renal protective role in glucose metabolism (Fig. 7**M**).

The role of metabolic regulation and energy reprogramming has been demonstrated in the progression of CKD (Miguel et al., 2025). We previously demonstrated that BYF significantly activated OXPHOS, which was reflected in the improvement of mitochondrial structural and functional homeostasis, increased the content of mitochondrial respiration complexes, and enhanced ATP production (Fig. 7). Activation of FAO or OXPHOS was significantly impaired in the CKD state, contributing to a shift in mitochondrial substrate dependency from fatty acids to glucose. Correspondingly, we also found that BYF altered the energy metabolism profile by inhibiting glycolysis activation (reduced ECAR) and enhancing FAO/OXPHOS (improved OCR) at the cellular level (Fig. 8**D, E**). Our findings suggest that BYF could be a promising therapy as a metabolic modulator, influencing glucose, lipid, and mitochondrial metabolism to combat fibrosis and CKD progression.

Finally, a compounds-targets network was constructed based on the predicted targets of BYF's core components, intersecting them with the BYF-reversed DEGs. The important targets exerted by the core components of BYF were selected (**Fig. S2**) and molecular docking was combined to predict binding affinity, aiming to understand whether these potential targets and components can interact effectively at the

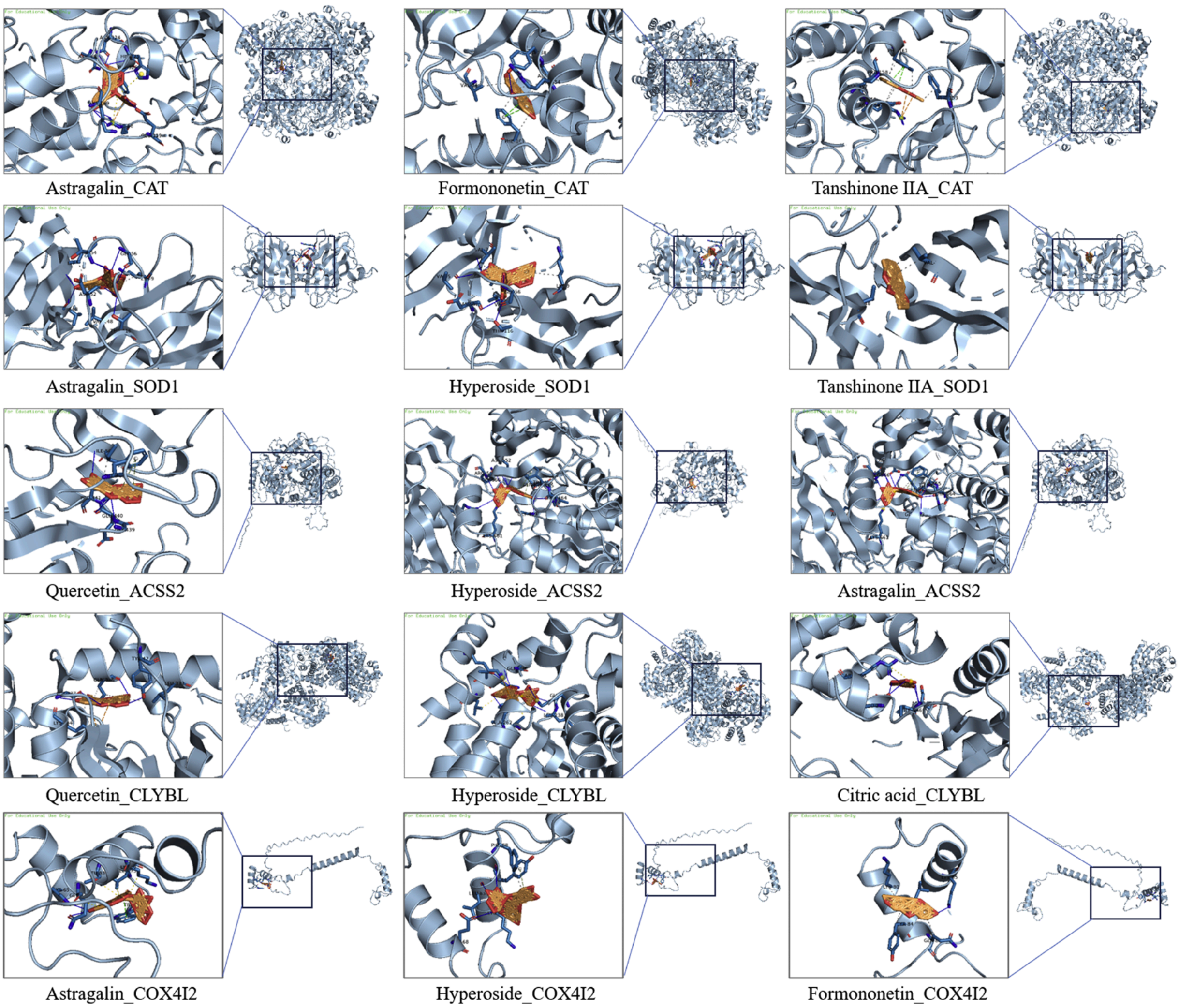

Fig. 9. Network analysis and molecular docking of BYF core component and predicted targets.

molecular level. The results showed that BYF core components, such as Astragalin, Quercetin, Formononetin, Hyperoside, Tanshinone IIA, and Citric acid, which have been fully elucidated in their protection against renal fibrosis (Lu et al., 2018; Sharma et al., 2020; Shi et al., 2024; Xu et al., 2020), have strong binding effects on key enzymes of peroxisomal antioxidation and mitochondrial OXPHOS, as well as key genes involved in the TCA cycle and ETC processes (Fig. 9). Thus, the association of BYF's renal protective effects under CKD conditions and its improvement of renal energy metabolism was further verified.

There are still limitations in this study. Future studies should focus on providing structural information and deconvolution of lipid isomers. As technology advances, new perspectives will emerge for identifying the specific lipid profiles affected by BYF in CKD. In the present study, we aimed to investigate the general effects of BYF on renal energy metabolism, including FAO, mitochondrial OXPHOS, and glucose metabolism. However, the interactions between lipid metabolism, glucose metabolism, and mitochondrial OXPHOS remain to be further explored. Future studies could adopt metabolic flux analysis to assess dynamic alterations in renal energy substrates during CKD progression.

Conclusion

In summary, our preliminary study indicated that BYF protected against CKD, likely by rebalancing renal energy homeostasis through the restoration of dysregulated FAO, glucose metabolism, and improving mitochondrial OXPHOS. Therefore, BYF holds promise as a therapeutic approach to slow the progression of kidney fibrosis inherent to CKD.

Supplementary figure legends

Figure S1. Lipidomic analyses of the kidneys in CKD rats treated with BYF. (A) Quality control base peak chromatogram in positive ion mode. (B) Quality control base peak chromatogram in negative ion mode.

Figure S2. System pharmacology analyses of BYF and identification of key targets associated with renal energy metabolism. (A) Venn diagram of targets identified by UHPLC-MS components and BYF-reversed DEGs. (B) Construction of network analysis of protein targets and BYF's core compounds.

CRediT authorship contribution statement

Bingran Liu: Writing – original draft. **Guowei Chen:** Writing – original draft. **Haoyu Mo:** Formal analysis, Data curation. **Xiaolin Liang:** Validation, Methodology. **Xiaoyan Su:** Formal analysis, Methodology. **Fuhua Lu:** Supervision, Project administration. **Qizhan Lin:** Supervision. **Xusheng Liu:** Supervision, Funding acquisition. **Jiankun Deng:** Conceptualization, Funding acquisition, Supervision. **Difei Zhang:** Conceptualization, Writing – review & editing, Funding acquisition.

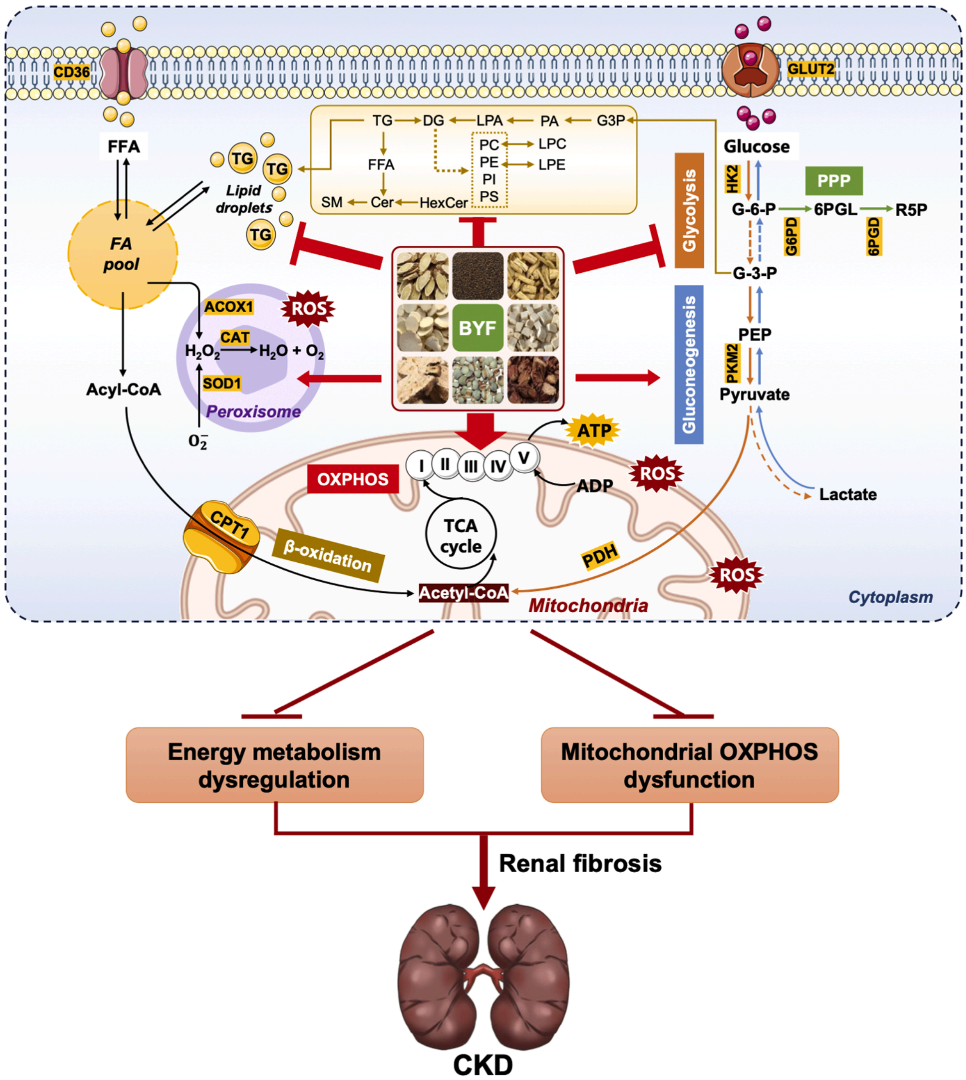

Fig. 10. Schematic diagram of the proposed mechanism underlying renal energy improvement exerted by BYF.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Scientific Research Projects of the State Key Laboratory of Traditional Chinese Medicine Syndrome (QZ2023ZZ26, QZ2023ZZ36, SKLKY2025C0007), the National Natural Science Foundation of China (82205106, 82174308 and 82405257), the China Postdoctoral Science Foundation (2023M740778), the Guangzhou Basic and Applied Basic Research Foundation (2023A03J0234), the Science and Technology Research Fund from Guangdong provincial hospital of Chinese medicine, China (YN2024GZRPY080), and the Administration of Traditional Chinese Medicine of Guangdong Province, China (20241139). We also extend our sincere gratitude to Dr. Yanmin Liu for their support regarding systems pharmacology and molecular docking analysis in this study.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.phymed.2025.157055.

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